Phase 3 – Security Monitoring, Threat Detection & Incident Response (SOC / SIEM)

Company Size: **300 Users**

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1 Solution Overview

Phase 3 extends the Zero Trust architecture implemented in Phase 1 and Phase 2 by introducing **centralized security monitoring, threat detection, and incident response capabilities**.

This phase implements a **Security Information and Event Management (SIEM)** platform to collect, correlate, and analyze security telemetry from all security layers deployed in the environment.

The primary platform deployed is:

- Microsoft Sentinel

Security telemetry is aggregated from:

• Identity systems
• Endpoint protection platforms
• Device management systems
• Application access services
• Microsoft 365 workloads

This enables:

✓ Centralized threat visibility
✓ Real-time attack detection
✓ Automated security response
✓ Security analytics and threat hunting

The result is a **Security Operations Center (SOC)-ready environment**.

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2 Security Architecture Extension

Phase 3 adds the **Monitoring and Detection layer** to the existing Zero Trust architecture.

Security Layers Covered:

Layer	**Technology**
Identity	Microsoft Entra ID
Device Management	Microsoft Intune
Endpoint Security	Microsoft Defender for Endpoint
CASB Monitoring	Microsoft Defender for Cloud Apps
Collaboration & Data	Microsoft 365

Layer	Technology
Security Analytics	Microsoft Sentinel

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3 Security Monitoring Objectives

The SIEM implementation aims to achieve the following objectives:

• Centralize security logs across all platforms
• Detect suspicious behavior patterns
• Identify compromised accounts or devices
• Investigate security incidents
• Automate response actions
• Improve security visibility for administrators

This transforms the architecture into a **proactive security model instead of reactive monitoring**.

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4 Log Source Integration

Security telemetry is ingested into Sentinel from multiple sources.

Identity Logs

From:

Microsoft Entra ID

Logs collected:

• Sign-in logs
• Risky sign-in logs
• User risk detection events
• Audit logs
• Privileged role activation events

These logs allow detection of:

• Credential attacks
• Impossible travel sign-ins
• Suspicious authentication attempts

Endpoint Security Logs

From:

Microsoft Defender for Endpoint

Telemetry collected:

• Malware detections
• Suspicious process behavior

• Endpoint compromise indicators
• Device vulnerability alerts

This enables early detection of:

• Ransomware attacks
• Credential dumping attempts
• Lateral movement activity

Device Compliance Logs

From:

Microsoft Intune

Collected data includes:

• Device compliance status
• Device enrollment events
• Device security configuration changes

This helps identify:

• Non-compliant devices attempting access
• Unauthorized device enrollments

CASB Activity Monitoring

From:

Microsoft Defender for Cloud Apps

Security events collected:

• Suspicious SharePoint file downloads
• Abnormal user activity
• External sharing attempts
• Cloud data exfiltration attempts

Microsoft 365 Activity Logs

From:

Microsoft 365

Monitored events:

• SharePoint access logs
• Exchange mailbox activity
• File downloads and deletions
• Administrative actions

These logs assist in detecting:

• Insider threats
• Unauthorized data access

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5 Threat Detection Rules

Microsoft Sentinel analytics rules detect suspicious security activity automatically.

Examples of detection rules implemented:

Impossible Travel Detection

Triggered when:

A user signs in from two geographically distant locations within a short timeframe.

Potential cause:

• Stolen credentials

Privileged Role Abuse Detection

Alert generated when:

• Unexpected activation of Global Administrator role
• Privileged role activated outside business hours

Source:

Microsoft Entra Privileged Identity Management

Suspicious File Download Detection

Alert triggered when:

• Large volumes of SharePoint files downloaded rapidly
• Sensitive documents accessed unusually

Malware Outbreak Detection

Triggered when:

Multiple endpoints detect malicious activity simultaneously.

Source:

Microsoft Defender for Endpoint

High-Risk Sign-In Detection

Generated when:

• Sign-in from anonymous IP addresses
• TOR network access
• Known malicious IP address

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6 Incident Response Automation (SOAR)

Security incidents can trigger automated response actions using **Sentinel Playbooks**.

Automation examples:

• Automatically disable compromised user accounts
• Force password reset
• Isolate infected endpoint devices
• Notify security administrators

These automated responses significantly reduce **Mean Time To Respond (MTTR)**.

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7 Security Dashboard & Visualization

Security dashboards are created in Sentinel to provide real-time visibility.

Dashboards display:

• Authentication trends
• Security alerts by severity
• Endpoint threat statistics
• Conditional Access policy activity
• Application access patterns

Administrators gain a **centralized security operations view**.

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8 Threat Hunting Capabilities

Security analysts can perform proactive threat hunting using Sentinel query language.

Investigations include:

• Searching suspicious login patterns
• Detecting lateral movement attempts
• Investigating abnormal data access
• Identifying compromised endpoints

This helps detect **stealthy attacks before major damage occurs**.

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9 Incident Investigation Workflow

When a security alert occurs:

1. Sentinel generates alert
2. Incident created automatically
3. Security analyst investigates event
4. Correlates logs from multiple systems
5. Determines attack scope
6. Executes remediation actions
7. Documents root cause
8. Updates security policies if required

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10 Security Operations Integration

Phase 3 enables the organization to operate a **Security Operations Center (SOC) model**.

SOC activities include:

• Continuous security monitoring
• Incident investigation
• Threat intelligence analysis
• Security policy improvements
• Security reporting

This ensures ongoing protection of enterprise resources.

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11 End-to-End Security Visibility

With Sentinel deployed, security visibility now spans across:

Security Domain	**Monitoring Enabled**
Identity	Yes
Devices	Yes
Endpoints	Yes
Applications	Yes
Data Access	Yes
Privileged Access	Yes

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12 Security Benefits

✓ Centralized log aggregation
✓ Early threat detection
✓ Faster incident response
✓ Reduced attack dwell time
✓ Improved forensic investigation capability
✓ Enhanced compliance monitoring

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13 Security Architecture Completion

After Phase 3, the organization now has a **fully mature Zero Trust security architecture**.

Phase	Security Capability
Phase 1	Identity, Device, and Data Protection
Phase 2	Secure Enterprise Application Access
Phase 3	Threat Detection, Monitoring, and Incident Response

🔐 Final Security Architecture Stack

Security Layer	Technology
Identity	Microsoft Entra ID
Privileged Access	Microsoft Entra Privileged Identity Management
Device Management	Microsoft Intune
Endpoint Protection	Microsoft Defender for Endpoint
Application Security	Microsoft Entra Application Proxy
CASB	Microsoft Defender for Cloud Apps
SIEM & SOC	Microsoft Sentinel

Final Outcome

The organization now operates a **complete Zero Trust security architecture** that:

• Verifies identity continuously
• Enforces device trust
• Protects corporate data
• Secures enterprise applications
• Detects and responds to cyber threats

This architecture is scalable for organizations growing from **300 users to several thousand users** while maintaining strong security governance.